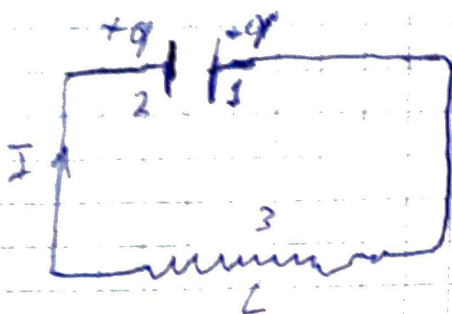




$$I = \frac{dq}{dt} = \dot{q}$$

$$\frac{dI}{dt} = \ddot{q}$$

Закон ОМА!

$$IR = \varphi_2 + \mathcal{E}_{12}$$

$$R=0$$

$$\varphi_1 - \varphi_2 = -\frac{q}{C}$$

$$\mathcal{E}_{12} = \mathcal{E}_5 = -L \left(\frac{dI}{dt} \right)$$

$$0 = \frac{-q}{C} - L \left(\frac{dI}{dt} \right) \quad \text{англ. и рус.}$$

$$q'' + \frac{1}{LC} q = 0 \quad \checkmark$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$q'' + \omega_0^2 q = 0$$

$$q = q_m \cos(\omega_0 t + \alpha) \quad \text{— облучение}$$

$$u = \frac{q_m}{C} \cos(\omega_0 t + \alpha) = U_m \cos(\omega_0 t + \alpha)$$

$$u(t) = \frac{q(t)}{C}$$

$$I = -\omega_0 q_m \sin(\omega_0 t + \alpha) = I_m \cos\left(\omega_0 t + \alpha + \frac{\pi}{2}\right)$$

$$\left(\frac{d}{dt} q_m \cos(\omega_0 t + \alpha) = -q_m \omega_0 \sin(\omega_0 t + \alpha) \right)$$

$$I_m = q_m \cdot \omega_0$$

$$U_m = \sqrt{\frac{L}{C}} I_m$$

формула Томсона

$$T = \frac{1}{\nu} = \frac{2\pi}{\omega_0} = 2\pi \sqrt{LC}$$

$$T = 2\pi \sqrt{LC}$$

Период АЧС равен?